



National University of
Science and Technology
POLITEHNICA Bucharest

NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY POLITEHNICA BUCHAREST



Faculty of Automatic
Control and Computers

CLEF 2026 · JENA, GERMANY · JOKER LAB · TASK 1 ENGLISH · TEAM VANGUARD

IROH: Insightful Ranking Of Humor using Multi-Stage Hybrid Retrieval with Rationale-Distilled LLM Judges

Ana-Maria Luisa Mocanu, Sebastian Mocanu, Ciprian-Octavian Truică, Elena-Simona Apostol

National University POLITEHNICA Bucharest | Academy of Romanian Scientists

{ana_maria.mogoase, sebastian.mocanu, ciprian.truica, elena.apostol}@upb.ro



Overview

- 01 Introduction
- 02 Our Contributions
- 03 System Architecture & Approach
- 04 Rationale Distillation
- 05 Results
- 06 Ablations
- 07 Limitations & Conclusions

Introduction

Humor Is Hard to Retrieve

Humor is ambiguous, culturally loaded, ironic, or carried by tone alone. Standard retrieval pipelines fail when both topical relevance and humor are required.

Prior Work

JOKER 2024 and 2025 relied on post-hoc filtering or zero-shot LLM classification.

System	JOKER Edition	MAP
Best English run, Task 1	2024	0.12
Best English run (Qwen filter-explainer)	2025	0.3501
IROH (ours), Task 1 English	2026	0.6347

Introduction

Meaning Plus Mechanism

Given a natural-language topic, the system must retrieve jokes, puns, and wordplay from a balanced corpus of humorous and non-humorous texts.

Humor Retrieval System

Humor Index

(jokes, memes, funny texts,
with semantic representations)



Retrieve & Rank

based on relevance
and humor signals



Top Retrieved Results

1



"Why don't cats play poker? Because they're always afraid of cheating."

2



"What do you call a cat who wears sunglasses? A cool cat."

3



"Why did the cat sit in the box? Because it wanted to be a purr-box."

Our goal: Team VANGUARD built IROH, a three-stage hybrid retrieval pipeline pairing sparse-dense retrieval and cross-encoder reranking with a LoRA-adapted LLM judge ensemble trained by rationale distillation.

Our Contributions

C1. Rationale Distillation Under Two Prompt Strategies. A lightweight generic pipeline and a structured typed pipeline distill query-aware rationales from Gemma 4; the simpler generic variant outperforms the richer typed one.

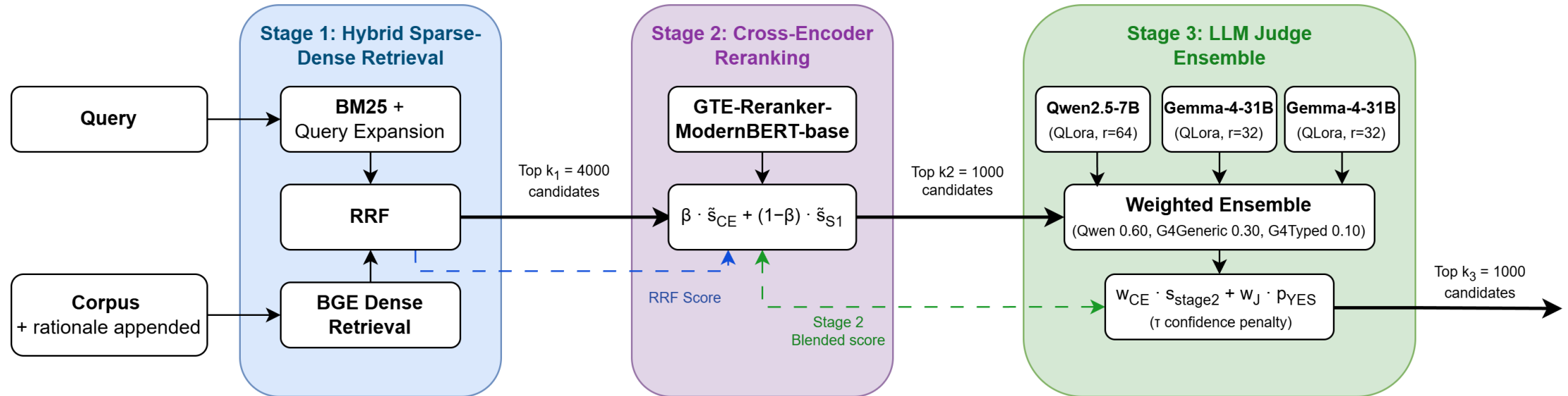
C2. Empirical Study of Structured Hard Negatives. Augmentation inflates local-validation scores while degrading official performance in nearly every configuration, an effect we attribute to distribution shift.

C3. Extensive Ablation Across Components. Model calibration and training-data composition matter at least as much as raw model capacity; the lighter model matches or beats its larger counterpart.

C4. Pipeline Configuration Analysis. Candidate-pool size governs recall depth, the CE/judge blend weight is a secondary knob, and the judge is the dominant top-rank signal.

System Architecture & Approach

Three-Stage Retrieval Architecture



IROH: hybrid sparse-dense retrieval, then cross-encoder reranking, then an LLM judge ensemble, scores carry forward across stages via weighted linear interpolation of min-max normalised signals.

Rationale Distillation

Because LLMs struggle to learn humor from labels alone, Gemma 4 generates a one-sentence, query-aware rationale for each query-document pair, both strategies below mirror positive and negative rationales for symmetric supervision.

Example , Tom Swifty: *“I bought myself fifty hamburgers, and I've only ten left,” said Tom with fortitude.*

Gemma 4's rationale: the pun arises from misapplying “fortitude” (courage) to a quantitative statement.

Generic Strategy ✓ Wins

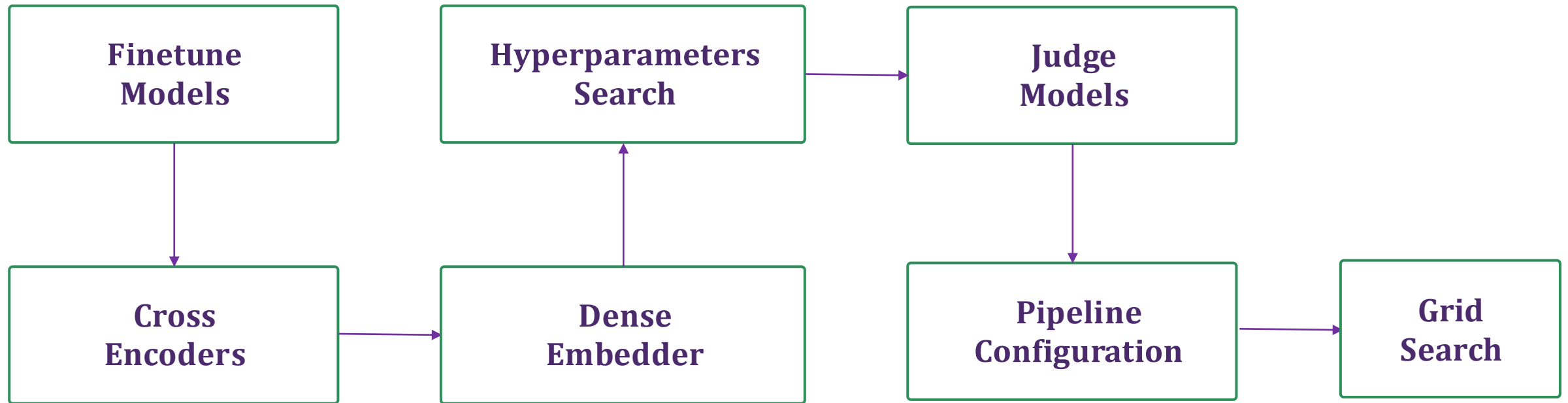
Uses a “General Wordplay” placeholder as query context for every example, keeping prompts lightweight. Generates three hard-negative types: literal rewrites, defused jokes, and wrong-topic jokes.

Typed Strategy

Retrieves the actual query text and enumerates seven humor mechanisms.

Results & Ablations

Steps of the Ablation



Ablation: Retrieval Components

Cross-Encoder Comparison

Model	Aug.	MAP
MiniLM-L-6	True	0.1793
MiniLM-L-6	False	0.2215
BGE-Reranker-Base	True	0.1229
BGE-Reranker-Base	False	0.0733
GTE-Reranker-ModernBERT-Base	True	0.2597
GTE-Reranker-ModernBERT-Base	False	0.2843

Dense Embedder Comparison

Model	Dim.	MAP
bge-base-en-v1.5	768	0.2843
bge-large-en-v1.5	1024	0.2823
bge-m3	1024	0.2634
bge-en-icl	4096	0.3290

Augmentation degrades the two strongest cross-encoder backbones, so we adopt GTE-Reranker-ModernBERT-Base without augmentation for Stage 2. bge-en-icl wins in isolation at 4096 dimensions, but at far higher cost (a 7B-parameter embedder); the smaller bge-base-en-v1.5 achieves higher MAP once paired with the full judge pipeline, so we select it.

Ablation: Judge Models & Final Configuration

Judge Model Comparison Across Data Variants

Model	Data	LoRA r	MAP
Qwen2.5-7B	typed	64	0.4933
Qwen2.5-7B	typed + aug	64	0.2987
Qwen2.5-7B	generic	64	0.6055
Qwen2.5-7B	generic + aug	64	0.4994
Gemma-4-31B	typed	32	0.5740
Gemma-4-31B	typed + aug	32	0.4538
Gemma-4-31B	generic	32	0.5718
Gemma-4-31B	generic + aug	32	0.4267

Final Submission Configuration

Knob	Value
Dense embedder	bge-base-en-v1.5
Cross-encoder	GTE-Reranker-ModernBERT (no aug.)
Pool sizes	k ₁ =4000, k ₂ =1000, k ₃ =1000
CE / judge blend	0.30 / 0.70
Ensemble weights	Qwen .60, G4-generic .30, G4-typed .10
Penalty	τ=0.30, λ=0.85 (no effect)
Final Score	0.6347

The smaller Qwen2.5-7B on generic rationales achieves the best single-judge MAP (0.6055), ahead of every Gemma-4-31B configuration, though Gemma is the stronger model per-variant on typed rationales.

Augmentation consistently degrades performance and generic rationales beat typed for both judges.

Limitations and Next Steps

Limitations

- **English-only.** Prompts, the humor mechanism taxonomy, and all finetuned components are English-only; transfer to other JOKER languages needs retraining.
- **Binary judge.** The YES/NO verdict forces a hard decision on what is arguably a relevance spectrum.
- **Rationale ceiling.** Rationales are accurate for clear wordplay but turn generic for culturally loaded or multi-layered humor.
- **Compute cost.** The full pipeline needs hours across multiple GPUs; submission limits forced OFAT sweeps over principled grid search.

Next Steps for IROH

- Extend the pipeline and humor taxonomy to other JOKER Task 1 languages
- Train the judge on graded relevance judgments for better-calibrated outputs
- Improve rationale generation for culturally loaded, multi-layered humor

Conclusions

What IROH Established

- A three-stage hybrid retrieval pipeline, sparse-dense retrieval, cross-encoder reranking, and a rationale-distilled LLM judge ensemble, for humor-aware document retrieval.
- **0.6347 MAP**, a 1.8× improvement over the best 2025 English run.
- The LLM judge is the dominant ranking signal; lighter, better-calibrated models (Qwen2.5-7B) and simpler generic rationales outperform larger or more structured alternatives.
- Structured hard-negative augmentation and richer typed rationales look better locally but degrade official performance, a distribution-shift finding reported openly.

Thank you! Questions?

